



JERUSALEM COLLEGE OF ENGINEERING, CHENNAI – 600 100
(An Autonomous Institution Affiliated to Anna University, Chennai)

Reg. No. :

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Question Paper Code : 221315

B.E./B.Tech. DEGREE END SEMESTER EXAMINATIONS, APRIL/MAY 2026

Fifth Semester

Computer Science and Engineering (Artificial Intelligence and Machine Learning)

JAL1501/JAL2501 – MACHINE LEARNING

(Common to Fifth Semester Computer Science and Engineering (Cyber Security) and Artificial Intelligence and Data Science, Sixth Semester Computer Science and Business Systems, and Seventh Semester Computer Science and Engineering)

(Regulation 2021/2023)

[CO: Course Outcome, BTL: Bloom's Taxonomy Level, M: Mark]

Time : Three hours

Maximum : 100 marks

Answer ALL questions

PART A – (10 × 2 = 20 marks)

	CO	BTL	M
1. Given vectors $u = [4, 2]$ and $v = [3, 6]$, find $u + v$ and uv .	1	2	2
2. What is inductive bias? Mention a few examples in different ML algorithms.	1	2	2
3. Define regression algorithms.	2	1	2
4. Write the advantages of batch gradient descent.	2	2	2
5. Define ensemble learning. What are its advantages?	3	2	2
6. What are Gaussian mixture models?	3	1	2
7. Define activation function.	4	1	2
8. What is hyperparameter tuning?	4	1	2
9. Mention the list of things that may have to be tracked in designing ML experiments.	5	2	2
10. What is a t-test?	5	1	2

PART B – (5 × 16 = 80 marks)

11. (a) How does Probably Approximately Correct learning contribute to machine learning? Elaborate with details. 1 3 16

Or

11. (b) Define bias and variance in the context of machine learning, and elaborate on the significance of the bias–variance trade-off with suitable illustrations. 1 3 16

12. (a) (i) Explain in detail about Linear Regression and its applications. 2 3 8

- (ii) Write a note on Least Squares Error Method. 2 2 8

Or

12. (b) (i) Write a brief note on Naïve Bayes classifier. 2 2 8

- (ii) Using Naïve Bayes classifier, predict the class for the given test data X = (age = youth, income = medium, student = yes, credit rating = fair). 2 3 8

id	Age	Income	Student	Credit Rating	Buys Computer
1	youth	high	no	fair	no
2	youth	high	no	excellent	no
3	middle_age	high	no	fair	yes
4	senior	medium	no	fair	yes
5	senior	low	yes	fair	yes
6	senior	low	yes	excellent	no
7	middle_age	low	yes	strong	yes
8	youth	medium	no	weak	no
9	youth	low	yes	weak	yes
10	senior	medium	yes	weak	yes
11	youth	medium	yes	strong	yes
12	middle_age	medium	no	strong	yes
13	middle_age	high	yes	weak	yes
14	senior	medium	no	strong	no

13. (a) (i) What are bagging and boosting? Explain with relevant examples. 3 3 8

- (ii) Outline the steps in AdaBoost algorithm with an example. 3 3 8

Or

13. (b) (i) Explain K-means algorithm and how it is used for clustering. 3 3 8

- (ii) Cluster the following set of data using K-means algorithm with initial value of objects 2 and 5 with the coordinate values (4, 6) and (12, 4) as initial seeds. 3 4 8

Objects	X – coordinate	Y- coordinate
1	2	4
2	4	6
3	6	8
4	10	4
5	12	4

14. (a) What is gradient descent optimization? Explain the stochastic gradient descent optimization method. 4 3 16

Or

14. (b) Develop a backpropagation algorithm for Multilayer Feed-Forward Neural Network consisting of one input layer, one hidden layer and output layer. 4 6 16

15. (a) Discuss in detail about the K-fold CV method with the algorithm and necessary illustrations. 5 3 16

Or

15. (b) For the following Confusion Matrix compute Accuracy, Precision, Recall, and F1-Score. 5 3 16

	Reality : Disease	Reality : No Disease
Predicted : Disease	25	5
Predicted : No Disease	15	55